

STYLE-CONDITIONED VIDEO CAPTIONING

TextSink

Four voices. One truth.

FORMAL

SARCASTIC

TECH-HUMOR

EVERYDAY-HUMOR

Grounded on frames + audio, styled by Gemma 4 — faithful and on-tone.

AMD Developer Hackathon: ACT II · Track 2 | Gemma 4 · Fireworks AI · AMD

Two jobs. Most entries do one.

Track 2 scores every caption on accuracy AND tone — across four different styles.

Accurate but flat

Frame-only bots describe the scene correctly, then miss the joke — and miss everything that was spoken.

Funny but wrong

Prompt-only bots write a punchy line — that invents objects, people, or events never shown on screen.

TextSink is built to win both axes at once.

One clip in. Four voices out.

A single 30s–2min clip becomes four captions — each faithful to the same facts.

● FORMAL

Neutral, precise, broadcast register.

● SARCASTIC

Dry and deadpan — never mean.

● TECH-HUMOR

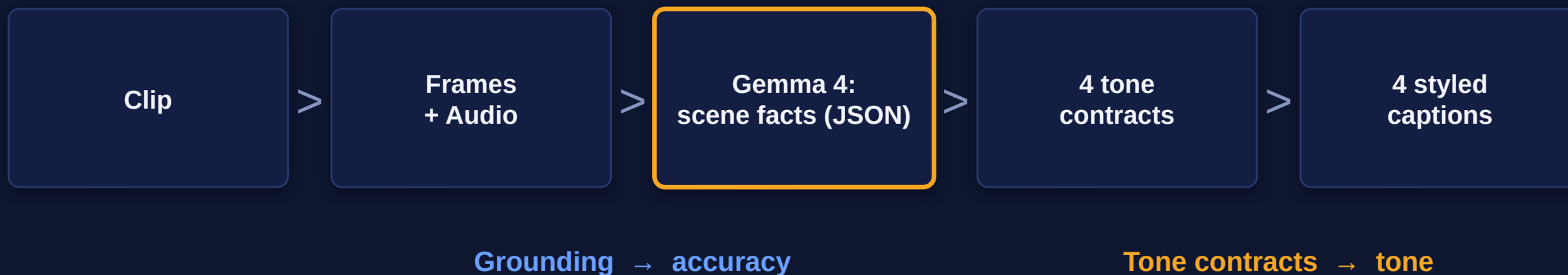
A joke a developer actually gets.

● EVERYDAY-HUMOR

Relatable, warm, zero jargon.

How it works

Separate the two things the judge scores — then optimize each.



One model — Gemma 4 — does the grounding AND the four captions. ffmpeg + Whisper feed it; per-style contracts shape the voice.

Accuracy: ground before you write.

- Sample frames every ~5s with ffmpeg (8–16 per clip).
- Pull audio into a transcript with Whisper — catches spoken content frame-only bots miss.
- Gemma extracts only what is visible or audible, as structured facts.
- Every caption is locked to those facts. No invented objects, people, or brands.

scene facts (grounded)

```
{  
  "setting": "indoor room, daytime",  
  "subjects": ["a person", "a laptop"],  
  "actions": ["types", "smiles at  
              screen"],  
  "notable_audio": ["upbeat music"],  
  "mood": "focused"  
}
```

Tone: four contracts, four voices.

Same grounded scene. Four Gemma 4 captions — real, verified output.

SCENE red circle + blue square on white; on-screen text 'OPEN 24/7'

FORMAL

"A red circle and a blue square on a plain white background, alongside the text OPEN 24/7."

SARCASTIC

"Groundbreaking shapes and a commitment to being open all the time. Truly a revolution in graphic design."

TECH-HUMOR

"The UI is finally live, but someone forgot the dark-mode toggle. At least the uptime is solid."

EVERYDAY-HUMOR

"The red circle and blue square are clearly working the night shift — they are open 24/7."

Gemma 4 — end to end.

\$5,500

addressable with one project:

\$2,500 Track 2 + \$3,000 Best Use of Gemma
4

- One model, Gemma 4 (26B MoE), does BOTH the visual grounding and the four styled captions.
- Served on-demand via Fireworks (H200), scale-to-zero — only bills while captioning.
- MIT-licensed, single Docker container — clone, build, run.
- Qwen 3.7 Plus + gpt-oss serverless fallback for zero-setup dev.

Runs today — not a mockup.

- Gemma 4 deployment verified: accurate grounding + 4 clean, on-tone captions.
- Real ffmpeg frame sampling and audio extraction verified.
- Full pipeline runs end-to-end and emits valid 4-style JSON; schema checks pass.
- Ships a self-judge that scores accuracy + tone to tune prompts with data.

run it

```
docker run --rm \
  -e FIREWORKS_API_KEY=$KEY \
  -v $PWD/videos:/data/in \
  -v $PWD/out:/data/out \
  cinetone
```

Four voices. One truth.

`github.com/<your-handle>/amd-track2-captioner`